Quantification: 682_z08
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	12.3333	12	4.3589	count
ring_length_r30_px	headline	463.111	465	206.708	px
ring_length_r60_px	reported	1040.44	1006	482.87	px
owned_length_px	ownership-dependent	3064.27	2235.2	2372	px
edge_count	ownership-dependent	152.556	123	115.863	count
mean_edge_length_px	ownership-dependent	19.1891	19.6911	4.2775	px
perimeter_px	normalised	253.461	234.894	106.721	px
ring_area_r30_px2	normalised	9783.33	9374	2822.56	px^2
ring_area_r60_px2	normalised	23226.1	23858	6006.58	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9809	1	0.0441	unitless
ring_density_r30	normalised	0.0453968	0.0475983	0.0143176	px^-1
ring_density_r60	normalised	0.0431795	0.0428723	0.0151849	px^-1
roots_per_100px_perimeter	normalised	5.1608	5.3871	1.1999	per 100 px
ring_attached_length_r30_px	network v2	423.889	430	203.989	px
ring_attached_length_r60_px	network v2	899.444	792	505.738	px
ring_length_w_r30_px	network v2	524.861	517.605	231.013	px
ring_length_w_r60_px	network v2	1178.34	1133.71	538.752	px
sholl_crossings_r10	network v2	11.1111	10	5.4416	count
sholl_crossings_r20	network v2	13.5556	14	5.615	count
sholl_crossings_r30	network v2	15	14	7.1239	count
loop_count	loops, report only	15.6667	12	13.6748	count
loop_edge_share	loops, report only	0.3324	0.3902	0.1577	share
ring_width_mean_r30_px	width	7.1358	7.3348	0.6168	px

Per field
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measure	value	units
analysed_area_px2	1.027e+06	px^2
total_skeleton_length_px	35837	px
canalicular_length_density_per_px	0.0348947	px^-1
junction_count	842	count
junction_density_per_px2	0.00081986	px^-2
lacuna_count	10	count

lacunae_per_px2	9.74e-06	px^-2
skeleton_component_count	272	count
mean_component_length_px	131.754	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0392578	px^-1
field_density_without_flagged_per_px	0.036564	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1011	px
unattached_component_count	57	count
canalicular_length_density_attached_per_px	0.0348947	px^-1
field_length_density_with_unattached_per_px	0.0358791	px^-1
loop_count	176	count
loop_edge_share	0.3683	share
edge_count	1895	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5558	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	34469	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
2	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
3	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
4	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
5	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
6	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
7	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
8	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
9	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False
10								347.049	976.84	False

Network per lacuna

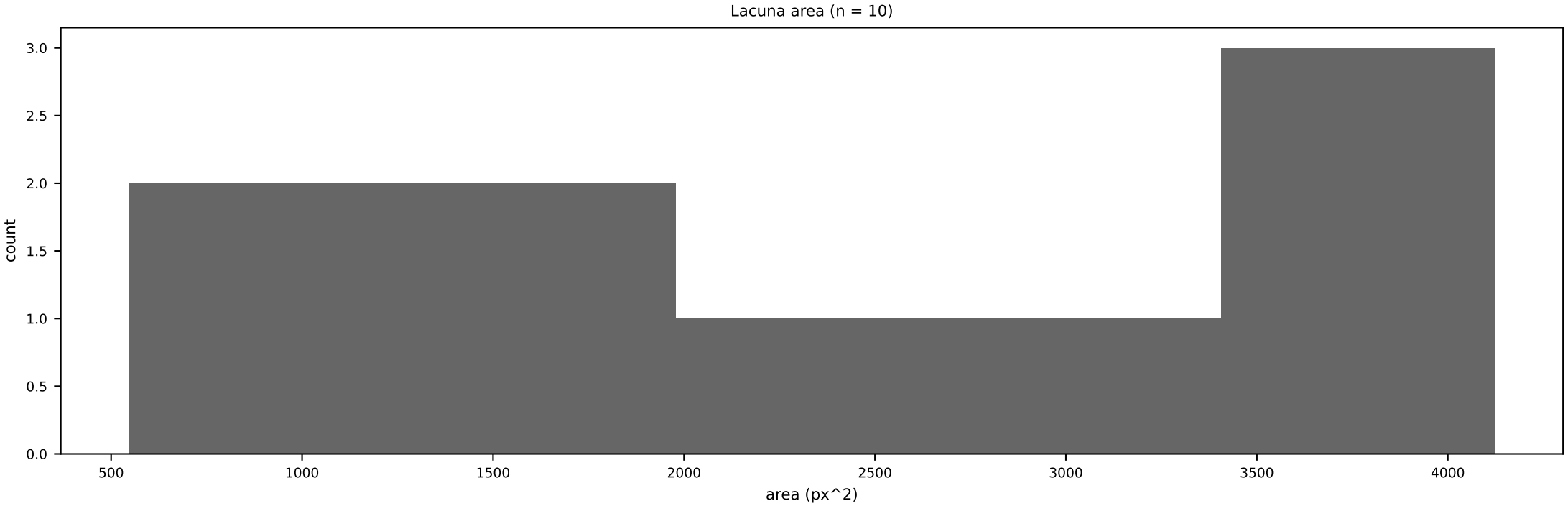
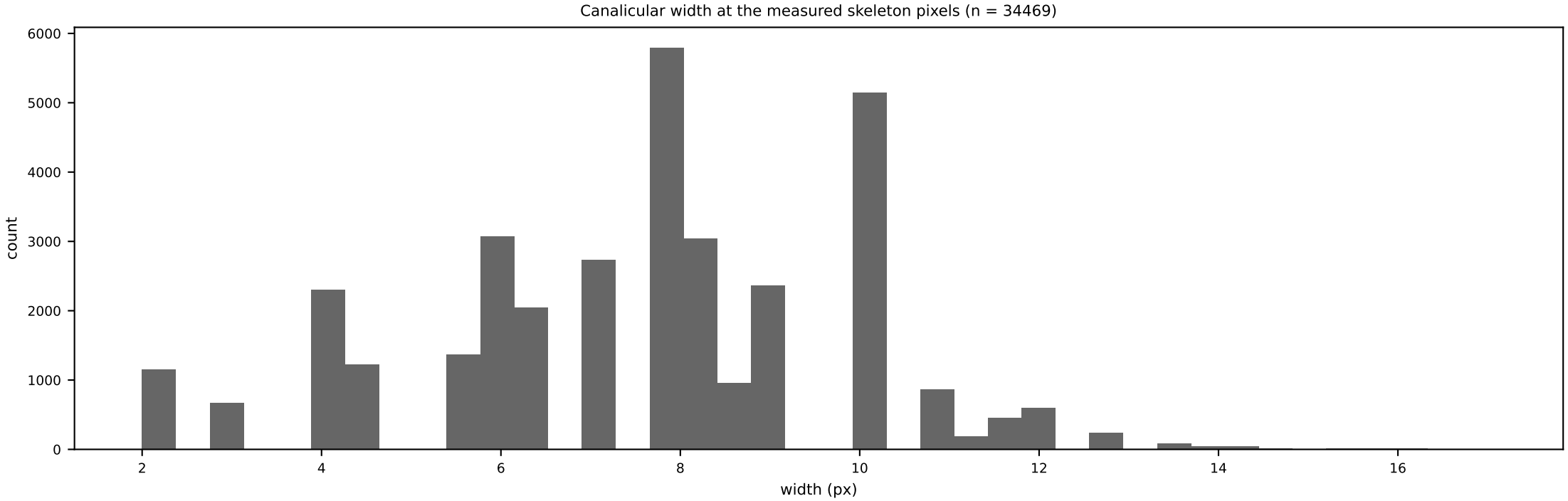
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	13	507	1199	3924.62	210	18.6887	7.8188
2	18	497	1135	3032.42	154	19.6911	7.4165
3	19	820	1896	5846.15	312	18.7377	7.4488
4	10	438	838	1629.95	75	21.7327	7.1556
5	9	465	1006	2235.2	123	18.1724	6.4611
6	4	176	436	425.357	12	35.4465	5.5304
7	15	651	1481	7603.14	347	21.9111	7.4475
8	12	379	846	1758.04	79	22.2537	7.3348
9	9	339	798	1478.78	65	22.7505	7.333

10	6	72	165	70.1127	8	8.7641	5.8058

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	234.894	9374	20769	1	0.9593	0.0540858	0.0577303	5.5344
2	313.948	11274	26474	1	1	0.0440837	0.0428723	5.7334
3	456.961	14983	32605	1	1	0.0547287	0.0581506	4.1579
4	233.238	9202	19621	1	1	0.0475983	0.0427093	4.2875
5	166.125	7387	17424	1	1	0.0629484	0.0577365	5.4176
6	122.604	3917	10590	0.6212	0.5802	0.0449324	0.0411709	3.2625
7	329.522	12020	29668	1	1	0.0541597	0.0499191	4.5521
8	222.752	9104	23858	1	1	0.0416301	0.0354598	5.3871
9	245.965	9693	25001	1	1	0.0349737	0.0319187	3.659
10	77.7401	5013	13615	1	0.8689	0.0143627	0.012119	7.718

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	492	1165	574.446	1346.42	12	14	17
2	478	979	567.659	1300.79	15	14	14
3	794	1835	915.097	2133.43	21	22	27
4	382	722	501.375	966.921	8	14	13
5	430	792	517.605	1133.71	10	13	13
6	161	370	205.894	515.571	5	6	7
7	580	1320	738.571	1645.3	15	18	22
8	309	666	437.446	985.703	9	14	15
9	287	553	392.777	911.122	8	12	13
10	63	63	78.7696	181.681	2	1	1

682_z08: distributions, pre-validation, px